Particulate Organic Matter Field Sampling Protocol

Version 1.1

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Protocol Revision Record

Version	Date Implemented	Description of Revisions
1.1	2023-02-27	Modified for GO-SHIP protocol
1.0		Initial Release

Acronyms, Abbreviations, Terminology

Term	Definition
POM	Particulate organic matter
POC/N	Particulate organic carbon/nitrogen
POP	Particulate organic phosphorus
PCOD	Particulate carbon oxygen demand
GF/F	Glass fiber filters
Inline/underway water	Sampling water that is pumped from beneath the ship to a receiving spigot in an onboard laboratory
Niskin bottle	Plastic cylindrical bottle used for collecting water samples at different depths.
CTD Niskin rosette water	Sampling water gathered at depth from the conductivity, temperature, depth (CTD) Niskin rosette
DI water	Deionized water from the Milli-Q system
GO-SHIP	Global Ocean Ship-Based Hydrographic Investigations Program
Bio-GO-SHIP	Biological GO-SHIP

Background

Summary

This protocol describes collection and filtration of marine POM samples using GF/F filters and can be adapted to collect water samples from individual Niskin bottles, CTD Niskin rosettes, or inline systems. This protocol is used by Bio-GO-SHIP Collaborators associated with the Martiny Lab at University of California, Irvine.

Method description and rationale

Triplicate discrete water samples, collected from an inline water system and/or CTD rosette, are filtered onto pre-combusted (500° C, 5 hours) 25mm glass fiber filters (GF/F) using an aspirator pump. Filters are frozen at -80° C until later analysis.

The recommended filtration volume for GO-SHIP samples is ~8 liters, which takes ~2 hours. Precautions are taken to minimize contamination between samples by rinsing collection bottles with sample water 3x prior to collection, and, once sample filtration is completed, rinsing collection bottles with DI and running DI water through the filtration system.

This measurement provides 1) total POP, POC/N, and PCOD 2) elemental stoichiometry, and 3) a pathway for creating optical proxies for particulate organic carbon.

Spatial coverage and environment(s) of relevance

This protocol can be used across any marine environment to effectively collect surface water samples for POM analysis.

Ship space requirements

Access to a sink with inline water supplied by a diaphragm or positive displacement pump. Approximately 8 ft length of countertop space, preferably near sink, for filtration system. A -80°C freezer is needed for storage of labeled GF/Fs.

Personnel Required

One person that is trained in filtering protocol prior to conducting on ship. Typically, two individuals that operate on 12 hour shifts.

Safety

There are no major safety concerns with this protocol. Gloves should always be worn, and eye protection should be worn when handling sodium sulfate.

Time needed to execute the procedure

The process of collecting triplicates of each sample type, filtering, and preparing filters for -80°C storage should take less than 2 hours.

Equipment

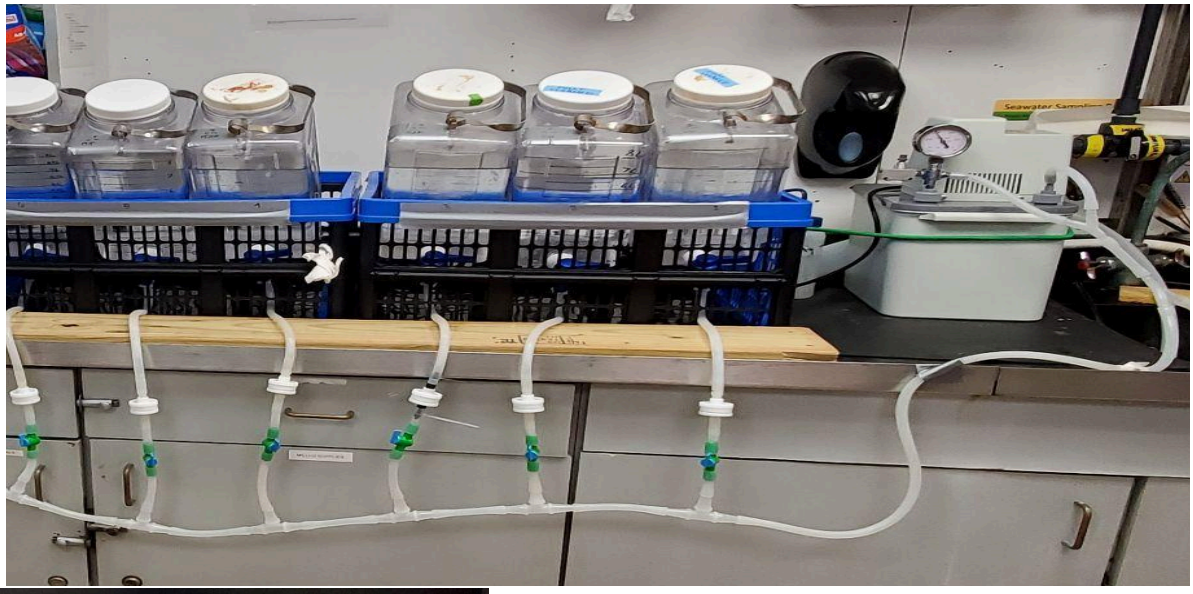
One application refers to the amount of each equipment item to facilitate triplicate samples of POC/N, POP, and PCOD. For research cruises, having additional spare/replacement equipment for each item is recommended.

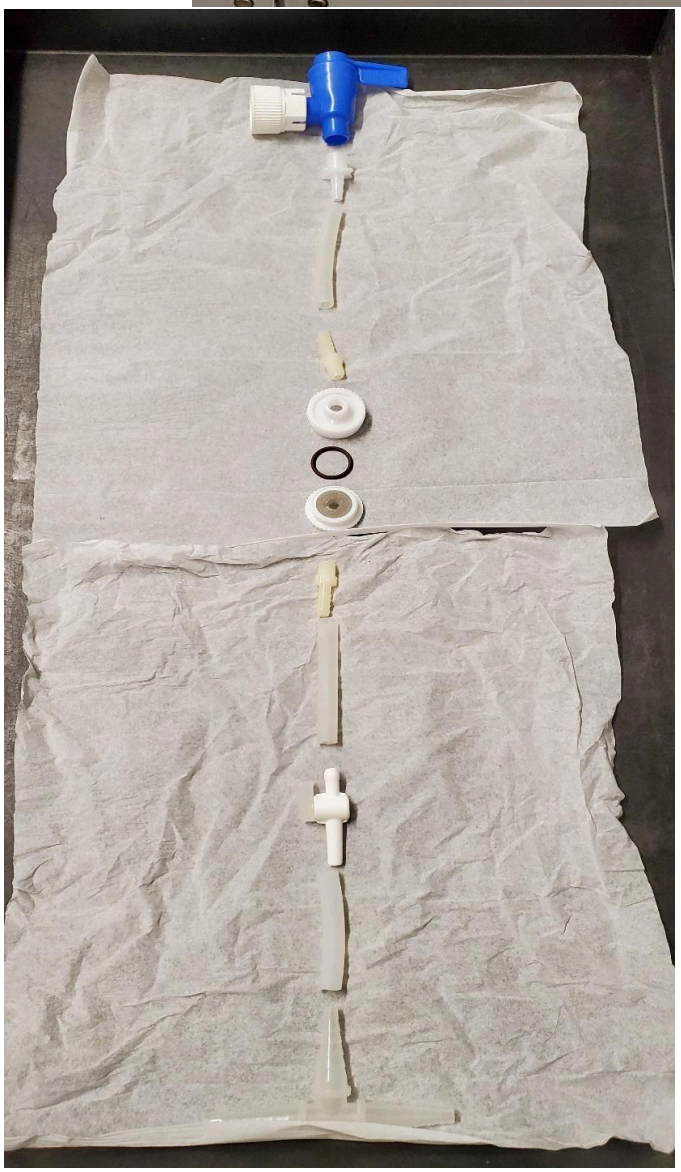
Description	Product Name and Model	Manufacturer	Quantity for one application	Remark
9L Nalgene carboy with spigot	Nalgene Rectangular Polycarbonate Clearboy Carboy with Spigot	Thermo Scientific	9	
Collapsable, plastic container			3	Each container can hold 3 carboys
Rebar			3	Allows for carboys to be held at an angle
Bungee cords			3+	Secures carboys at an angle, and extra to secure the filtration system as needed
Ratchet straps			3	Stabilizes the frame of the collapsable containers
Tubing adaptor			9	Connects rubber tubing to spigot
Rubber tubing			Several meters of tubing in various, precut lengths	Rubber tubing in various lengths is assembled for the filtration system, as described in the protocol
30µm nylon mesh			9 squares of mesh, roughly 10 cm x 10 cm	
Rubber bands			9	Secures mesh to tubing
Aspirator pump	Aspirator Pump, 18 L/min, 9.5l Bath, 115 VAC	Cole-Parmer	1	1 pump needed for protocol, at least 1 spare should be packed for research cruises
Glass Fiber Filters (GF/F)	Whatman 25 mm		9	Pre-combusted at 500° C for 5 hours, stored in pre-combusted aluminum foil envelopes
Filter cups			9	
O-rings			9	
Filter adaptor			18	Connects filter cups to tubing
Inline valve			9	
T-connector			9	
Sodium sulfate Anhydrous	CAS: 7757-82-6		5 mL of solution	24 g of sodium sulfate (pre-weighed and

				packed in flacon tubes) per 1 L of DI
1 L Nalgene bottle			1	Holds sodium sulfate solution
5 mL syringe			1	To apply sodium sulfate solution
50 mL falcon tube			2+	To hold 5 mL syringe and sodium sulfate solution, use for storage and organization as needed
50 mL falcon tube holder			1	To hold falcon tubes
Tweezers			1	To place and remove filters from frit, can be stored in a falcon tube
Absorbent underpads	Absorbent Underpads with Waterproof Moisture Barrier	VWR International	3+	Use and replace as needed to line area under carboys, etc.
Nitrile gloves			1 set for sample collection, 1 set for filtration process, more as needed	Gloves should be worn during sample collection and during the filtration process
Eye protection			1	To be worn when handling sodium sulfate
Pre-combusted foil envelopes			9	To store filters for long term storage
Pre-printed POM labels			9	To store filters for long term storage
Fine tipped sharpie			1	To fill in labels as needed
Quart and gallon ziplock bags, whirl-pak bags				Use as needed to store samples in freezer
Duct tape			1 roll	Use as needed for filtration setup
Eyelet hooks			12+	Use as needed for filtration setup
Parachute cord/rope			Several meters in length	Use as needed for filtration setup
1000 mL graduated cylinder				Use as needed in calculating filtration volumes
Wash bottle			1	Filled with DI water, use as needed for cleaning filtration system between samples
Plumbing tape, electrical tape, label tape			1 roll each	Use as needed for sealing spigots, setting up filtration system, etc.
Pens, mechanical pencils				Use as needed for sample log
Scissors				Use as needed for filtration setup

Standard Operating Procedure

Setup





Above: Filtration set-up for POM, including collapsible bins with rebar, nalgene spigoted carboys, filtration frits, tubing, connections, and aspirator pump. **Ratchet straps should be wrapped around the bins for stability.** Carboys sit on top of rebar. The aspirator basin has two overflow ports (facing the sink in photo). Place the aspirator close to the sink in case water needs to be drained from the bottom overflow spigot. Attach tubing to the upper overflow port so any overflow goes directly into the sink (below).

Left: Deconstructed set up of filtration from the carboy spigot to the T connector. Replace any parts if cracked, punctured, or continuously leaking.



Below: Aspirator pump set-up for use. Check tubing for cracks. When connecting the pump to tubing and carboys, you can use the pressure gauge as shown in top photo, or two tubing ports shown below.



Inline water collection, filtration, and storage

- Sampling typically occurs from underway water at 0600, 1200, and 2000 local time, and once per day from CTD bio-casts ideally close to solar noon.
- Turn on the inline system and run for less than a minute to flush the lines
- You need 3 9L nalgene spigoted carboys for each sample collection type (if collecting POP and POC/N, then you need 6 carboys)
- Rinse each carboy out 3x with sample water
- Fill each carboy to overflowing before putting on lid
    - Each carboy is labeled and lists empty and full masses, which is used to estimate volume filtered
- To prepare for filtration, prepare by putting 1 combusted GF/F on each filter frit, wet watch filter with a bit of DI
- Ensure brown o-ring is in place on top of GF/F
- Carefully tighten each frit to finger tight, do not damage filter edges. You can prep this within 1 hours prior time of water collection
- Glance over the filtration system and ensure that all valves are closed
- Place each carboy in the collapsible bins at a 45 degree angle, which prevents particle settling to the bottom
- Attach filter frit hose to the carboy spigot
- Check that aspirator pump has water in the basin to the fill line
- When ready, open each carboy spigot valve and each valve below the filter frit. Ensure water is flowing

- With all connections opened, turn on aspirator pump and check pressure if gauge is available
    - Do not exceed 5 Hg (125 mm Hg, 0.17 bar, -17 kPA) to avoid particle rupture
- Monitor filtration- should take less than 3 hours
    - If filtration is taking longer than 3 hours, talk with Adam or Star over whatsapp.
    - **If you do not filter the entire volume in carboy**, you will need to estimate volume filtered using a graduated cylinder or estimating with markings on the carboys.
- Close the valve below the filter frit just as the sample water runs dry
    - If not filtering full volume, close the carboy spigot valve, allow the remaining water to run dry, and then close the valve below the filter frit
- **For POP: after sample water has completed filtering, syringe 5 mL of Na_2SO_4 on each GF/F (rather than DI) to remove trace phosphorus on GF/F**
    - It is okay to reuse the syringe throughout entire cruise. You can store it in an empty falcon tube.
- Turn off aspirator when all samples are finished
- Gently remove each GF/F with forceps, holding the edge of the filter rather than the touching the sample area
- Fold GF/F in half with sample side inward, and envelop in pre-combusted foil
- Attach corresponding sample label with appropriate information filled in
- Store in -80°C freezer
- Fill each carboy with ~0.5-1L of DI water and turn on aspirator pump to run DI water through the lines. Turn off pump and close connections when done
- Empty the water from the aspirator basin and replace with sink water to minimize salt corrosion
- Fill out e-log
- Periodically clean the top of the aspirator case and other surfaces with sink water and a sponge to mitigate salt build-up. Unplug aspirator prior to cleaning
- Every 3-4 days, run diluted bleach (2-10%) through the system (use less than 0.5 L per carboy) and then flush well with DI water

CTD niskin collection

- Using water from niskins fired at the surface, attach tubing to the niskin outflow port/nipple
- Place 9L nalgene carboy without the lid below the niskin, with hose placed inside carboy
- Open the outflow port and fill carboy with less than 0.5L of sample water
- Quickly close port, place lid on carboy, slosh water around, and pour out onto the ground
- Repeat this rinsing process 2 more times
- Fill carboy to overflowing
- When rinsing carboy, be mindful of water needs (each carboy is ~9L when filled to overflowing). How much water does the niskin hold (often 10L)? Are there other water needs for that particular niskin? Do not use too much sample water when rinsing
- Follow same protocol for filtration for inline

Na_2SO_4 solution

- The sodium sulfate powder is pre-weighed in labeled 50 mL falcon tubes
- Pour one falcon tube of powder into a clean 1L nalgene bottle
- Fill bottle to 1 L and shake until powder is fully dissolved (may take a few minutes)
- Store in room temperature

Basic troubleshooting

- If the filter frit is leaking, close the carboy spigot port, unscrew the frit, ensure o-ring is in place, and retighten. Consider tightening more or less than previously, as over tightening or under tightening can both lead to leakage
- If there is leaking from carboy spigot, wipe down the inside of the spigot and the threading, and reattach. If leaking continues, replace spigot
- There may be leaking from other connections through the tubing. Try your best to locate exactly where and replace parts as needed
- If aspirator seems to be overheating, switch the power off and allow water in the basin to drain. Fill aspirator up to fill line with cold sink water and resume filtration. Use ice if available. You may need to leave the pump off for longer.
- If aspirator is constantly overheating or if it sounds significantly louder than normal when running, consider switching out pumps

Post-Cruise Sample Storage

- During cruise, filters should be stored in a -80 C freezer
- During final week or days of cruise, start prepping Techni-Ice sheets
    - How to use Techni Ice?
    - After hydrating, leave the sheet on towels or paper towels to dry out, and pat dry before putting in freezer
    - Do not put more than 1 newly hydrated techni-ice sheet in the freezer at a time
        - The freezer will drop temperature with each new sheet you add
        - All sheets should end up in the -80 freezer, but you can pre-freeze sheets in the -20 freezers if space is available and then move to the -80 once frozen
    - The cooler needs at least 1 sheet per 7 L of cooler space- please freeze more sheets than you may need so you can fill up all empty cooler space with ice sheets
    - Minimize any empty space in the cooler
    - Make sure the bottom of the cooler is lined with ice sheets
    - You can cut sheets into smaller sizes to assist in filling empty space- it is easier to cut them prior to hydration
    - Do not pack cooler until just prior to departing ship. If available, you can pre-chill the cooler by storing it in a walk-in fridge or freezer